

GDELT Crisis Forecasting

Deven Shah | MET CS 777 - Big Data Analytics | Term Project

Introduction

Organizations that respond to political violence work under a scheduling constraint. Humanitarian agencies pre-position supplies, peacekeeping missions reallocate patrols, newsrooms decide where to send correspondents, and analysts decide which desks to staff. All of these decisions have to be made before violence peaks, not after it. Even two or three weeks of warning changes what is operationally possible. The conflict-risk indices that are most widely used, however, are refreshed quarterly or annually and are built from structural indicators such as GDP per capita, regime type, and history of past war. Those indicators explain well why some countries are fragile, but they say very little about which particular week the fragility turns into events.

The GDELT 2.0 Event Database (gdeltproject.org) is an unusual data source in that it operates on exactly the timescale that the operational question needs. GDELT machine-codes news reports from global media in over 65 languages into structured event records using the CAMEO taxonomy, capturing who did what to whom, where, and when. Version 2.0 covers February 2015 to the present and is updated every 15 minutes; the earlier GDELT 1.0, which covers 1979 to 2015 and is not used here, is updated daily. The data is distributed free of charge as zipped, tab-separated CSV files with 61 columns, one file per 15-minute window, which works out to 96 files per day and hundreds of millions of event records in total. Two properties of the data shape everything that follows. It is continuously updated, which makes weekly forecasting feasible, and it is derived from news coverage rather than from verified incident reports, which means it measures reporting about the world rather than the world itself.

This project asks a single question: can we predict whether a country-week will see a spike in material conflict one to four weeks out, using only open news-derived event data? A spike is defined relative to each country's own recent history rather than in absolute terms. Specifically, a country-week is a spike when its count of material conflict events is at least twice its trailing eight-week average and at least eight events in absolute terms. The relative definition is a deliberate choice. Absolute event counts are dominated by a small number of chronically violent countries and by how much media attention a region happens to attract, so a model trained to predict absolute counts would mostly learn a fixed ranking of countries. The operationally interesting question is escalation: which countries are about to become more violent than they usually are.

To answer it, a PySpark pipeline aggregates the GDELT event stream into country-week feature vectors, labels each week against the spike definition above, and trains a Gradient-Boosted Trees classifier from Spark MLlib to forecast the label one to four weeks ahead. The models are evaluated on a strict chronological hold-out against two baselines: logistic regression, and a persistence rule that simply alerts whenever a country is already spiking. Over 9.1 million events were aggregated into 15,930 country-week rows covering 118 countries and 135 weeks, each described by 42 features. Five findings emerge. First, open news data does carry a real forecasting signal, but a modest one: PR-AUC of 0.172 at a one-week horizon against a base rate of 0.090, roughly a 1.9x lift over random, decaying to 1.46x at four weeks. Second, the model beats the persistence rule at every horizon, and the margin widens from 35 percent to 99 percent in relative F1 as the horizon lengthens. Third, the predicted probabilities are well calibrated across most of their range, which makes them usable for triage rather than only for ranking. Fourth, the escalation-ladder hypothesis, that hostile words precede hostile deeds, receives partial support: trailing verbal conflict is consistently among the most important features, but the country's own material conflict history dominates. Fifth, and least expected, Gradient-Boosted Trees do not uniformly outperform logistic regression; which model looks better depends on which metric is used, and the reason for that is itself informative. Alongside the pipeline, a Flask web application serves the current risk scores, the evaluation results, and per-country histories.

Methodology

Data acquisition and sampling

GDELT 2.0 publishes 96 export files per day. Downloading all of them for a multi-year window would mean roughly 89,000 HTTP requests, which is well beyond what is needed to answer the research question on a laptop. The downloader therefore fetches a uniform subsample of 8 of the 96 files per day, evenly spaced across the 24-hour cycle. Uniform spacing rather than random selection matters here: it guarantees that every day contributes the same set of 15-minute slots, so the diurnal rhythm of news publication is sampled identically in every week. Because the

features and the label both depend on week-over-week relative volume rather than on absolute counts, a consistent 1-in-12 subsample preserves the quantity being measured.

The download covers 2024-01-01 through 2026-07-22, which amounts to 7,472 export files and 85.6 MB of compressed data. Of the 61 columns in the GDELT export schema, 11 are retained at download time:

GLOBALEVENTID, SQLDATE, EventRootCode, QuadClass, GoldsteinScale, NumMentions, NumSources, NumArticles, AvgTone, ActionGeo_CountryCode, and DATEADDED. Filtered rows are appended to one gzipped CSV per month, and a manifest records which 15-minute files have already been processed so that the download is resumable and incremental.

Cleaning

Three cleaning steps run before aggregation. Duplicate records are dropped on GLOBALEVENTID, since the same event can appear in more than one export window. Events are then filtered on staleness: an event is kept only when the difference between DATEADDED, the moment it entered the stream, and SQLDATE, the date the event is reported to have occurred, is between zero and seven days. This filter is important and easy to overlook. GDELT continually codes retrospective and anniversary coverage, so an article published in 2026 about a massacre in 1994 produces an event record dated 1994. Without the staleness filter those records would be counted into historical weeks, smearing the weekly signal with material that is not news. Finally, each event is assigned to a country using ActionGeo_CountryCode, the FIPS code of the place where the event occurred, and to a week using the Monday of the week containing its reported date.

Aggregation and grid completion

Cleaned events are grouped by country and week. Each group produces a total event count; counts for each of the four CAMEO quad classes (verbal cooperation, material cooperation, verbal conflict, material conflict); counts for each CAMEO root code from 10 to 20, which form an escalation ladder running from demand, disapprove, reject, threaten, protest, force posture, reduce relations, and coerce through assault, fight, and mass violence; the average Goldstein score and average tone; and the summed mention, article, and source volumes.

The aggregation is then joined onto a complete country-by-week grid built from the cross product of every observed country and every week in the range. This step is a correctness requirement rather than a cosmetic one. Without it, a week in which a country generated no coded events is a missing row rather than a row of zeros, and every subsequent lag and rolling window silently steps over the gap, so a nominal one-week lag would sometimes reach back two or three weeks. Count columns are filled with zero and the two average columns with a neutral 0.0.

Countries averaging fewer than 40 sampled events per week across the whole period, roughly 480 unsampled events, are then dropped, because a ratio-based spike test is meaningless when the weekly counts are in the single digits. This leaves 118 countries.

Feature engineering

Each country-week row carries 42 features in two groups. The 24 level features describe the week itself: total events, the four quad-class counts, the eleven root-code counts, average Goldstein score, average tone, summed mentions, articles and sources, and three share ratios (verbal conflict share, material conflict share, cooperation share). The 18 dynamics features describe how the week compares with the country's recent past: material conflict lagged one through four weeks, verbal conflict lagged one and two weeks, one-week lags of protest count, tone, Goldstein score and total events, the trailing eight-week mean and standard deviation of material conflict, the trailing eight-week mean of verbal conflict, a spike ratio and a z-score of the current material conflict count against that trailing baseline, two four-week trend differences, and a log-transformed event total.

Every rolling statistic is computed over a window that ends at the previous week and excludes the current row. That constraint is what makes the exercise a forecast rather than a description: the feature vector for week t contains nothing observed after week t , and the label attached to it is the spike flag of week $t+h$, obtained by leading the flag forward within each country. A model that scores well is therefore genuinely predicting, not reading the answer out of a contemporaneous feature.

Label definition

For each country-week the spike threshold is the greater of eight events and twice the trailing eight-week mean of material conflict. The week is a spike when its material conflict count reaches that threshold. The absolute floor of eight events exists to stop tiny denominators from producing false positives: a country whose baseline is 0.5 events

per week would otherwise be flagged for recording two, which is noise rather than escalation. Weeks with fewer than four weeks of prior history receive a null flag rather than a zero, so a thin baseline never masquerades as a quiet one. Under this definition 8.89 percent of the 15,458 country-weeks with a valid flag are spikes. The forecast target label_h for week t is the spike flag of week $t+h$, for h in 1, 2, 3, and 4.

Train and test split

The split is chronological and includes a gap. The distinct weeks are ordered, the first 75 percent form the training period and the remainder the test period, and a gap of h weeks is inserted between them. The gap is what makes the split honest at longer horizons: without it, a training row from the last week before the cutoff would carry a label describing a week that falls inside the test period, leaking the answer across the boundary. Cutoffs range from 2025-11-24 for the one-week horizon to 2025-11-10 for the four-week horizon, giving between 10,738 and 11,328 training rows and between 3,776 and 3,894 test rows depending on the horizon. Positive rates are 8.6 to 8.8 percent in training and 9.0 percent in test, so the two periods are comparably difficult.

Models

The proposed model is a Spark MLlib GBTCClassifier with 80 trees of depth 4, step size 0.1, 80 percent row subsampling, a square-root feature subset per split, and a minimum of 10 instances per node, fed by a VectorAssembler over the 42 features and seeded for reproducibility. Gradient-Boosted Trees suit this problem because the features are tabular and wildly different in scale (raw counts in the thousands alongside share ratios between 0 and 1), the relationships are not linear, and the model reports feature importances that can be inspected. Two baselines establish what the model has to beat. Logistic regression, fitted on the same features after mean and variance standardization with L2 regularization, tests whether the additional model complexity is doing any work. A persistence rule, which alerts for week $t+h$ whenever week t is already a spike, tests whether the model is doing anything beyond restating the present. The persistence rule has no parameters and no training cost, so it is the genuine floor. No class weighting or resampling is applied to either probabilistic model, which keeps the predicted probabilities interpretable as probabilities and makes the calibration analysis meaningful.

Evaluation metrics

Accuracy is deliberately not reported. With a 9 percent positive rate, a model that always predicts no spike scores 91 percent accuracy while being entirely useless, so accuracy would obscure rather than reveal. The headline metric is PR-AUC, computed as step-wise average precision, whose random-model floor is the positive rate itself, here 0.090. ROC-AUC is reported alongside it. Precision, recall, and F1 are reported at an operating threshold chosen to maximize F1 on the training predictions only and then applied unchanged to the test set, so the test data never influences its own operating point. Recall at a fixed 10 percent false alarm rate answers the practical question directly: if an analyst can tolerate alerts on one in ten quiet country-weeks, what fraction of real spikes does the system catch? Finally a calibration curve over ten quantile bins compares mean predicted probability against observed spike rate.

Implementation and runtime

The pipeline is implemented in PySpark 3.5 and runs on a laptop with a local[*] SparkSession, a 4 GB driver, and 16 shuffle partitions. All row-level work happens in Spark; only the aggregated country-week table is ever collected to the driver. On the reference run, building features from 7,472 raw files took about 12 minutes and training plus evaluating all three models across all four horizons took 59 seconds. The disparity is the point: the expensive stage is the distributed scan over millions of raw events, and the modeling stage operates on a table whose size is bounded by the calendar.

Results

All results below come from the pipeline run of 2026-07-22 over GDELT data from 2024-01-01 to 2026-07-22. Table 1 summarizes the data that reached the models.

Table 1. Dataset after acquisition, cleaning, and filtering.

Property	Value
Source	GDELT 2.0 Event Database, 15-minute export files
Date range	2024-01-01 to 2026-07-22

Property	Value
Sampling rate	8 of 96 daily export files (uniform 1-in-12)
Export files downloaded	7,472
Columns retained	11 of 61
Compressed size on disk	85.6 MB
Events aggregated after cleaning	9,123,051
Material conflict events	1,296,063 (14.2%)
Countries after volume filter	118
Weeks covered	135 (2023-12-25 to 2026-07-20)
Country-week rows	15,930
Rows with a valid spike flag	15,458
Overall spike rate	8.89%
Features per row	42

Forecasting performance

Table 2 gives the full comparison. Every figure is computed on the held-out test period only.

Table 2. Test-set performance by lead time. GBT = Gradient-Boosted Trees, LR = logistic regression. The random-model PR-AUC floor is the test positive rate, 0.090.

Lead	Model	PR-AUC	ROC-AUC	Recall @10% FAR	Precision	Recall	F1
+1w	GBT	0.172	0.700	0.249	0.197	0.312	0.242
+1w	LR	0.192	0.693	0.235	0.146	0.493	0.225
+1w	Persistence	-	-	-	0.177	0.181	0.179
+2w	GBT	0.144	0.666	0.203	0.161	0.264	0.200
+2w	LR	0.133	0.648	0.178	0.126	0.691	0.213
+2w	Persistence	-	-	-	0.134	0.135	0.134
+3w	GBT	0.144	0.665	0.183	0.154	0.204	0.175
+3w	LR	0.143	0.654	0.198	0.135	0.587	0.220
+3w	Persistence	-	-	-	0.088	0.089	0.088
+4w	GBT	0.131	0.658	0.162	0.146	0.263	0.188
+4w	LR	0.133	0.653	0.159	0.139	0.581	0.224
+4w	Persistence	-	-	-	0.100	0.100	0.100

Three patterns stand out. The signal is real: at a one-week horizon the GBT reaches a PR-AUC of 0.172 against a random floor of 0.090, a lift of roughly 1.9x, and a recall of 0.249 at a 10 percent false alarm rate, which is about 2.5x what random alerting would produce at the same alarm budget. The signal decays with horizon as expected, falling to a PR-AUC of 0.131 at four weeks, a lift of 1.46x. And the model clearly beats persistence at every horizon: F1 is 35 percent higher at one week, 49 percent at two, 99 percent at three, and 87 percent at four. The widening margin is the more meaningful comparison. Persistence collapses beyond a week or two because the fact that a country is spiking now says progressively less about whether it will spike three weeks from now, whereas the model retains a usable fraction of its accuracy.

Gradient-Boosted Trees versus logistic regression

The comparison against logistic regression is less clear-cut than expected, and reporting it honestly requires looking at what the F1 column is actually rewarding. Logistic regression records the higher PR-AUC at one week (0.192 versus 0.172) and the higher F1 at three of the four horizons, yet the GBT has the higher ROC-AUC at all four horizons and the higher recall at a fixed 10 percent false alarm rate at three of four. Table 3 resolves the apparent contradiction by showing how many alerts each model actually issues at its F1-maximizing operating point.

Table 3. Alert volume at the F1-maximizing operating point chosen on training data. Logistic regression buys its recall by alerting on a large share of all country-weeks.

Lead	Test rows	Actual spikes	GBT alerts	LR alerts	GBT precision	LR precision
+1w	3,894	349	553 (14.2%)	1,178 (30.3%)	0.197	0.146
+2w	3,894	349	573 (14.7%)	1,910 (49.0%)	0.161	0.126
+3w	3,776	339	449 (11.9%)	1,474 (39.0%)	0.154	0.135
+4w	3,776	339	608 (16.1%)	1,421 (37.6%)	0.146	0.139

At the two-week horizon, logistic regression achieves its higher F1 by raising an alert on 1,910 of 3,894 country-weeks, or 49 percent of the world, in order to capture 241 of 349 spikes at a precision of 0.126, barely above the 9 percent base rate. The GBT alerts on 573 country-weeks, 14.7 percent, at a precision of 0.161. An alerting system that flags half the world every week is not usable regardless of what its F1 says, so F1 at an unconstrained threshold is a poor summary for this problem. Under a fixed alert budget, which is how such a system would actually be operated, the GBT is ahead at three of four horizons. The accurate conclusion is that the added model complexity purchased a better-ordered head of the ranking rather than a uniformly better model, and that logistic regression's average precision at one week is genuinely competitive and should not be explained away.

Calibration

Table 4 compares predicted probability against observed spike frequency for the one-week GBT, binned into deciles of predicted risk.

Table 4. Calibration of the +1 week GBT on the test set, ten quantile bins.

Decile	Mean predicted	Observed rate	Country-weeks
1 (lowest)	0.020	0.010	390
2	0.029	0.031	389
3	0.043	0.039	389
4	0.056	0.074	390
5	0.069	0.064	389
6	0.086	0.075	389
7	0.110	0.103	390
8	0.135	0.141	389
9	0.166	0.159	389
10 (highest)	0.268	0.200	390

Across the first nine deciles predicted and observed rates track closely: 0.110 predicted against 0.103 observed, 0.135 against 0.141, 0.166 against 0.159. The probabilities can therefore be read as probabilities rather than as bare rankings, which is what makes them useful for triage. The exception is the top decile, where the model predicts 0.268 but the observed rate is 0.200. The highest-risk bucket is overconfident by about seven percentage points, so the most alarming scores should be discounted somewhat.

What the model learned

Table 5 lists the ten most important features at the shortest and longest horizons.

Table 5. Top ten GBT feature importances by lead time. Importances sum to 1 across all 42 features.

Rank	Feature (+1 week)	Imp.	Feature (+4 weeks)	Imp.
1	quad4_mean8 (trailing 8w material conflict)	0.076	quad4_lag3 (material conflict, 3w ago)	0.057
2	quad4_lag3 (material conflict, 3w ago)	0.061	quad4_lag1 (material conflict, 1w ago)	0.053

Rank	Feature (+1 week)	Imp.	Feature (+4 weeks)	Imp.
3	quad3_mean8 (trailing 8w verbal conflict)	0.046	quad4_lag4 (material conflict, 4w ago)	0.050
4	quad4_lag4 (material conflict, 4w ago)	0.045	quad4_lag2 (material conflict, 2w ago)	0.046
5	quad4_std8 (trailing 8w volatility)	0.037	quad4_mean8 (trailing 8w material conflict)	0.039
6	events_lag1 (total events, 1w ago)	0.036	quad3_mean8 (trailing 8w verbal conflict)	0.038
7	quad4_lag2 (material conflict, 2w ago)	0.035	events_lag1 (total events, 1w ago)	0.037
8	tone_avg (average article tone)	0.032	mentions_sum (mention volume)	0.037
9	quad4_spike_ratio (current vs baseline)	0.031	quad2_material_coop (material cooperation)	0.035
10	tone_lag1 (average tone, 1w ago)	0.030	quad4_material_conflict (current week)	0.032

The importances are notably diffuse. The single most important feature accounts for only 7.6 percent of total importance across 42 features, so no one signal drives the forecast. At the one-week horizon the model leans on the level and volatility of recent material conflict, together with trailing verbal conflict and article tone. At four weeks the trailing mean gives way to the individual weekly lags, and raw coverage volume enters the top ten through mention and source counts. The interpretation is that short-horizon forecasting depends on how much conflict a country has had recently and how erratic it has been, while longer-horizon forecasting depends more on the shape of the recent trajectory and on how much attention the country is attracting overall. Trailing verbal conflict appears third at one week and sixth at four weeks, a consistent presence at both ends.

Current risk scores

The trained models are applied to the most recent complete week, 2026-07-20, producing a probability of a spike at each horizon for all 118 countries. The highest scores at the one-week horizon were Djibouti at 0.688, South Sudan at 0.335, and Laos at 0.308. Djibouti is instructive: it was already in a spike at the time of scoring, with 9 material conflict events against a trailing baseline of 3.8, and the model largely re-expressed that fact. This is consistent with recent material conflict dominating the one-week feature importances, and it is a reminder that the shortest-horizon model overlaps substantially with the persistence baseline it outperforms.

Discussion

Does open news data carry a forecasting signal?

Yes, and the honest characterization is that it is real, modest, and decaying. A PR-AUC roughly 1.9 times the random floor at one week, falling to about 1.5 times at four weeks, is not an oracle. It is, however, considerably better than nothing, and it is achieved with no proprietary data, no human analysts, and no structural covariates. The appropriate framing for a system built on these numbers is triage. Given an analyst who can examine perhaps 15 percent of country-weeks closely, the model roughly doubles the number of genuine escalations that fall inside that review window compared with reviewing at random. It should not be framed as an alarm that fires when violence is coming, because at these precision levels most alerts will be false.

Why the model comparison depends on the metric

The finding that logistic regression is competitive on PR-AUC while losing on ROC-AUC and on recall at a fixed alert budget is worth dwelling on, because it is a general lesson about evaluating rare-event classifiers rather than a quirk of this dataset. F1 at an unconstrained threshold rewards a model for reaching high recall no matter how many alerts that requires, and logistic regression exploits exactly that by flagging between 30 and 49 percent of all country-weeks. PR-AUC integrates precision across the entire recall range, including regions where recall is high and precision has collapsed to near the base rate, which no operator would ever use. Recall at a fixed false alarm rate asks the question an operator actually faces, and by that measure the GBT is ahead at three of four horizons. The practical recommendation from this project is to select on recall at a fixed alert budget and to treat F1 as secondary whenever the positive class is rare.

The escalation-ladder hypothesis

The proposal predicted that verbal conflict events, threats and demands and denunciations, would spike before material conflict, and that this would be the principal source of predictive signal. The data supports a weaker version

of that claim. Trailing eight-week verbal conflict is consistently among the top six features at both the shortest and longest horizons, so verbal conflict does carry information about material conflict to come. But the strongest features throughout are the country's own material conflict history: its trailing mean, its volatility, and its individual weekly lags. The accurate statement is that most of what is predictable here is inertia, that conflict-prone countries continue to be conflict-prone, and that the ambient level of verbal conflict adds a real but secondary increment on top of it. What the results support is that the verbal conflict level is informative, not the sharper claim that a verbal spike reliably precedes a material spike.

Limitations

The most consequential limitation is that GDELT measures coverage rather than ground truth. Every count in this project is a count of machine-coded news reports, so a spike in coded material conflict may reflect a spike in attention rather than in violence: an election, a summit, a newly arrived foreign correspondent, or a single widely syndicated story can all raise counts. No independent incident dataset such as ACLED or UCDP was used to validate the labels, so the target is strictly a spike in reported material conflict. This should be stated plainly whenever the results are presented.

Four further limitations apply. The 1-in-12 subsample preserves relative week-over-week volume in expectation, but countries near the 40-events-per-week inclusion floor have noisy counts, and sampling variance alone can create or conceal a marginal spike; a full-firehose run would need the count-based thresholds rescaled by a factor of twelve. The relative spike definition means chronically violent countries require very large absolute jumps to trigger, so the model ranks escalation rather than absolute danger. Evaluation rests on a single chronological cutoff around November 2025 with only 339 to 349 positive test cases per horizon, which is enough to establish that the signal exists but not enough to place tight confidence intervals on any individual figure; walk-forward validation over multiple rolling origins would be the natural next step. Finally, countries are attributed solely by the location of the event, so actor-level and cross-border dynamics are invisible to the model, and the GBT hyperparameters were fixed a priori rather than searched, meaning the comparison is between two sensible defaults rather than two tuned optima.

Scaling beyond the laptop

The pipeline was deliberately structured so that scale is a matter of configuration rather than redesign. All row-level work, cleaning, aggregation, grid completion, and window functions, executes as Spark transformations, and only the aggregated country-week table is collected to the driver. That table's size is set by the calendar, 118 countries by 135 weeks, not by the sampling rate, so moving from 8 to all 96 files per day multiplies the raw scan by twelve while leaving the modeling stage essentially unchanged at roughly 20,000 rows. The full firehose over the same date range would be about 89,400 export files and around 1 GB compressed. Because the expensive stage is a single distributed scan, the same code scales by adding executors to the feature build; the project README documents both the single large instance path and the cluster path, along with the threshold rescaling the larger sample requires.

Future work

Four extensions follow directly from the limitations. Validating the labels against an independent incident dataset would establish how much of the predicted signal corresponds to violence rather than to coverage. Walk-forward validation across many rolling origins would replace the single split and yield uncertainty estimates. Adding actor-level features from the Actor1 and Actor2 country codes would let the model see dyadic escalation between specific pairs of states, which the current location-only attribution cannot represent. And a systematic hyperparameter search, together with sequence models that consume the weekly series directly rather than hand-built lags, would test whether the modest ceiling observed here is a property of the data or of the model class.

Conclusion

This project asked whether a spike in material conflict in a given country-week can be predicted one to four weeks in advance using only open news-derived event data. A PySpark pipeline aggregated 9.1 million GDELT 2.0 events into 15,930 country-week rows spanning 118 countries and 135 weeks, described each row with 42 strictly backward-looking features, labeled spikes relative to each country's own trailing eight-week baseline, and trained Gradient-Boosted Trees against a logistic regression baseline and a persistence rule on a chronological hold-out with a gap between train and test.

The answer is a qualified yes. At a one-week horizon the model reaches a PR-AUC of 0.172 against a 0.090 random floor and catches 24.9 percent of spikes while alerting on 10 percent of quiet country-weeks; performance decays smoothly to a PR-AUC of 0.131 at four weeks. The model beats the persistence rule at every horizon, and by a widening margin as the horizon grows, which is the clearest evidence that it is forecasting rather than restating the present. Its probabilities are well calibrated across the lower nine deciles and modestly overconfident in the highest, making them suitable for triage. The comparison with logistic regression proved to depend on the metric: the GBT wins on ROC-AUC at every horizon and on recall at a fixed alert budget at three of four, while logistic regression edges it on PR-AUC at one week by issuing several times as many alerts. Feature importances support a weaker form of the escalation-ladder hypothesis than proposed, with trailing verbal conflict consistently informative but the country's own material conflict history dominating.

The deliverables are the PySpark pipeline, a command-line interface covering download, feature construction, training, and scoring, a Flask web application that serves current risk scores and evaluation results without recomputing them, and a documented path to running the same code against the full GDELT firehose on a cluster. The broadest conclusion is a cautionary one worth stating alongside the numbers: open news data does contain a genuine early-warning signal at a one-to-four-week horizon, but it is a signal about reporting, it is modest in size, and it is most useful for prioritizing scarce analyst attention rather than for raising alarms.